

from X.509 certificates. In some cases, as noted, the limitations on the semantic interpretation of parts of a Matter certificate follow from limitations applied by [\[RFC 5280\]](#).

A certificate comprises a record of the following conceptual fields:

Certificate Text

- Version Number
- Serial Number
- Signature Algorithm ID
- Issuer Name
- Validity period
  - Not Before
  - Not After
- Subject name
- Subject Public Key Info
  - Public Key Algorithm
  - Subject Public Key
- Issuer Unique Identifier
- Subject Unique Identifier
- Extensions
- Certificate Signature Algorithm
- Certificate Signature

#### 6.5.1.1. ASN.1 Object Identifiers (OID)

Several important components of X.509 certificates follow the pattern commonly used in ASN.1 data models where some types are constructed with an ASN.1 object identifier (OID) to identify each variant. For example, the cryptographic algorithm used in the digital signature is identified by its OID.

Matter certificates do not use ASN.1 OIDs. Instead, each valid ASN.1 OID SHALL be mapped to a Matter TLV tag within its reference category. Each reference category defines the context of the Matter tag, and tag values are assigned to the reference categories according to the type of fields where they can appear in X.509 certificates.

#### 6.5.2. Matter certificate

A Matter certificate encodes a subset of the object identifiers (OIDs) specified in X.509. Only some attribute types for relative distinguished names are valid, only certain cryptographic algorithms (corresponding to the algorithms as defined in [Chapter 3, Cryptographic Primitives](#)) are used, and only a limited set of extensions are used. Therefore, every Matter certificate can be represented as a corresponding X.509 certificate. However, the converse is not true; not every X.509 certificate can be represented as a Matter certificate.

The signature included in a Matter certificate is the **signatureValue** of the corresponding X.509 certificate, not a signature of the preceding Matter TLV data in the Matter certificate structure. Accord-